

HS4002: Lecture 5A

Sampling

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**What Will Be On The Test in
Week 7?**

What will be assessed

- Course content from weeks 2 to week 6 (inclusive).
- Your knowledge of the pragmatics of conducting survey research.
- Your general mastery of statistical analysis: how, what, and why we calculate and interpret models, statistics and data visualizations.
- MCQs, structured questions, and a case study.

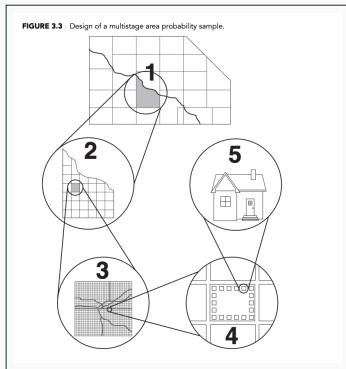
What will NOT appear

- Minutiae on programming:
 - e.g. How do you draw a histogram?
- Minutiae on statistics:
 - e.g. What is the formula for sample skewness?

Common Sampling Frames

Area Probability Sampling

Figure 1: Area Probability Sampling



We use area as a sampling frame.

Address-Based Sampling

- We can use a list of addresses as our sampling frame.
 - e.g. a list of all residential addresses.
- Limitations:
 - Coverage can be lower in certain areas.
 - Multiple-person dwellings may only have 1 address.
 - People with >1 address over-represented.

Random Digit Dialing (RDD)

- We can either obtain (or randomly generate) a list of all phone numbers in a given area/country.
- Limitations:
 - Contact rates can be low.
 - People are less cooperative.

- We can also use proprietary lists compiled by other organizations.
- Examples:
 - Organization membership lists

Types of Probability Sampling

What Is Probability Sampling?

Probability sampling refers to sampling methods where every member of the sampling frame is given a known, nonzero chance of being included in the sample.

Simple Random Sampling

Simple random sampling is when each member of the population is given an equal chance of being included, members are selected randomly, and their selection is independent of one another.

Cluster Sampling

- Sample units are grouped (or clustered) and then the groups are randomly sampled, rather than each unit sampled independently.
- Pros:
 - Reduces survey costs.
- Cons:
 - Increases sampling variance since units within a clusters have a lot in common.
 - This means, there is less unique information coming from the units drawn.

Stratified Sampling

- Sample units are divided into homogenous subpopulations (strata) before sampling.
- When done appropriately, can reduce sampling variance in measures of interest.
- Two types:
 - Proportionate sampling, where we sample sub-groups at equal rates.
 - Disproportionate sampling, where we sample at unequal rates.
- Can be combined with cluster sampling ("multistage sampling").

How Large Should My Sample Be?

How many?

An obvious point: you want your samples to be as large as possible, subject to constraints in resources.

Deciding “big enough”

Think about two things:

1. What is your desired level of confidence in your estimates?
2. What kinds of sample variation are you expecting?

$$se = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation, and n is the size of the sample.

$$\text{MoE} = z * \frac{s}{\sqrt{n}}$$

where z is the Z-score for the desired level of confidence, s is the sample standard deviation, and n is the size of the sample.

$$\text{CI} = (\bar{x} - z * \frac{s}{\sqrt{n}}, \bar{x} + z * \frac{s}{\sqrt{n}})$$

$$n = \frac{(z * s)^2}{\text{MoE}^2}$$

Survey Weights

- This weight corrects for any differences in sampled members' chances of being selected into the sample.
- It reflects the number of people represented by each completed interview.
 - It is the inverse of the overall probability of selection.

- We can also use known characteristics of a target population to adjust the weights of a sample such that it matches the target population.
 - Note that it does not make the sample representative for characteristics that are uncorrelated with the characteristics used for weighing.

Thanks for listening :)